

EEE241L Multisim Assignment

Electrical Circuits II Lab

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Section:	04	Date:	08/09/2026

Objective:

To simulate inductive and capacitive load circuits in NI Multisim, measure peak voltages and time differences using an oscilloscope, and calculate load current, phase angle, and power drawn by the load.

Circuit Schematics:

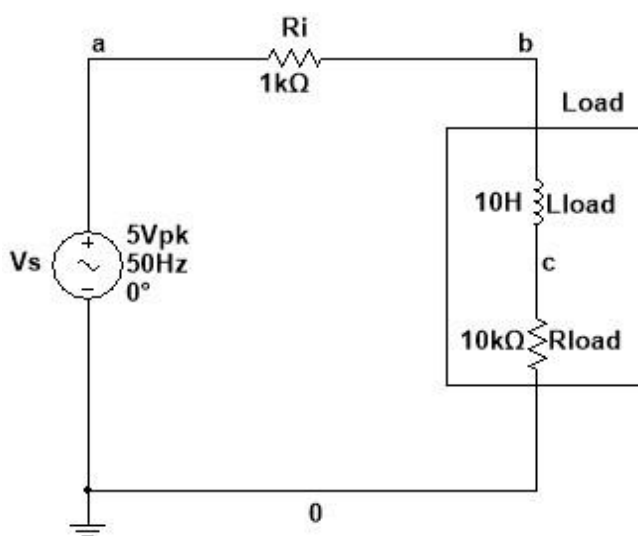


Figure 1: Inductive load in a series circuit

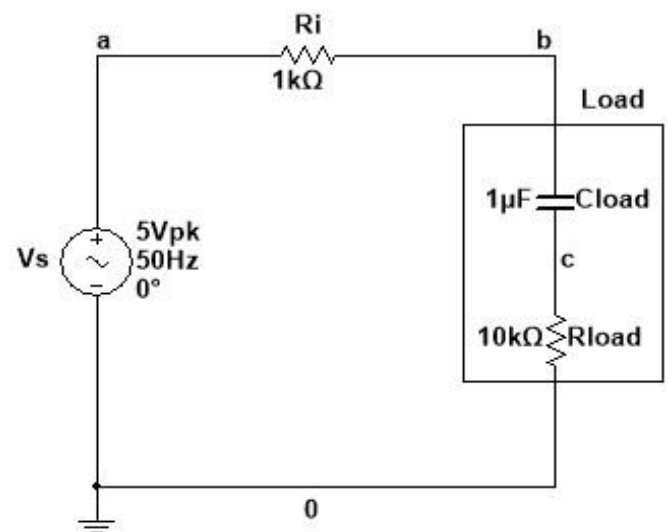


Figure 2: Capacitive load in a series circuit

Problem 1: Inductive Load in a Series Circuit

Simulate the inductive load circuit shown in Figure 1. Measure the load voltage, the voltage across R_i , and the required time differences using the oscilloscope. Then complete Table 1 and calculate the load current and power drawn by the load.

Table 1: Inductive Load

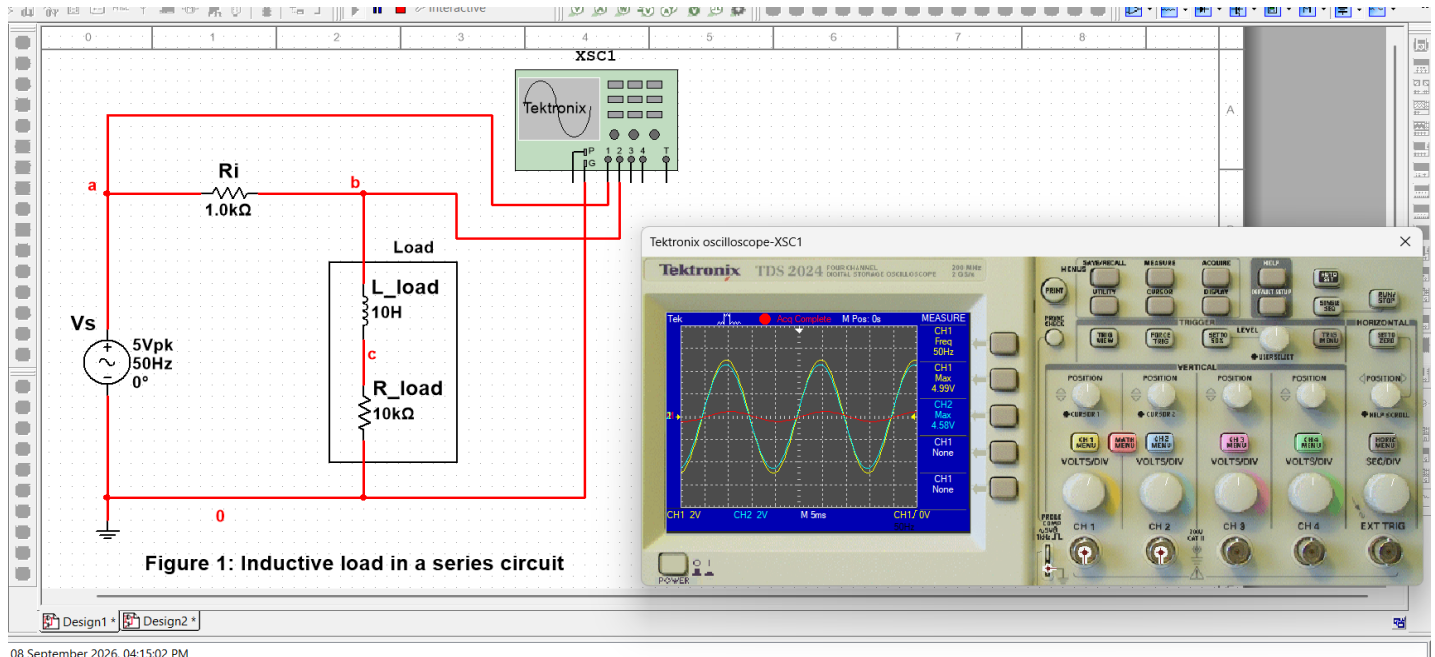
Reading	Parameter	Value
1	V_{Load} (peak)	4.58 V (peak)
2	Δt for load voltage	140 μS
3	θ for load voltage	$\theta = (\Delta t \times f \times 360) = (140 \times 10^{-6} \times 50 \times 360) = 2.52^\circ$ (Leading)
4	V_{R_i} (peak)	440 mV (peak)
5	Δt for R_i voltage	880 μS
6	θ for R_i voltage / current	$\theta = (\Delta t \times f \times 360) = (880 \times 10^{-6} \times 50 \times 360) = 15.84^\circ$ (Lagging)
7	$I_s \angle \theta$	$I_s = \frac{V(R_i)}{R_i} = \frac{440 \angle -15.84^\circ}{1000}$ $= 0.44 \text{ mA}_{\text{pk}} \angle -15.84^\circ$
8	Power drawn by the load, P	$P = \left(\frac{1}{2} I_{\text{pk}}^2 \times R_{\text{load}}\right) = 0.5 \times (0.44 \times 10^{-3})^2 \times 10^4$ $= 0.968 \text{ mW}$

Calculation (Inductive Load):

for Inductive load,
 θ for load voltage,
 $\theta = \Delta T \times f \times 360^\circ$
 $= 140 \times 10^{-6} \times 50 \times 360^\circ$
 $= 2.52^\circ$ (leading)
 θ for R_i voltage / current,
 $\theta = \Delta T \times f \times 360^\circ$
 $= 880 \times 10^{-6} \times 50 \times 360^\circ$
 $= 15.84^\circ$ (lagging)

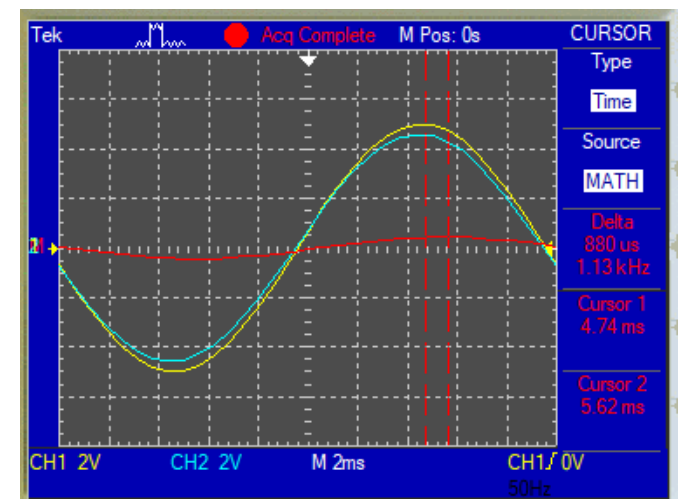
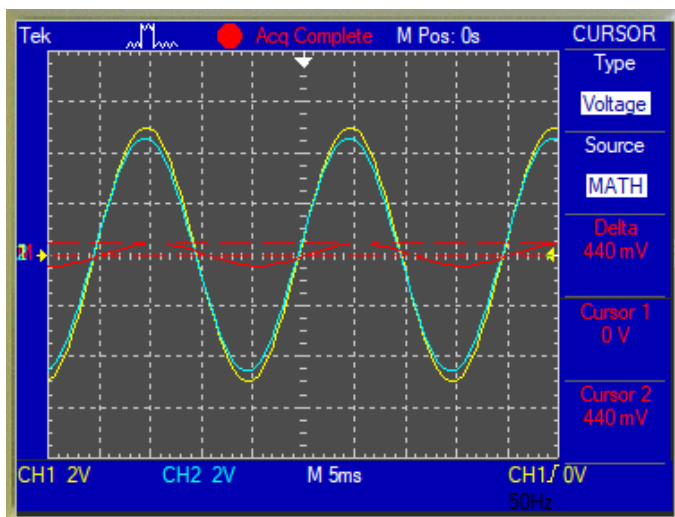
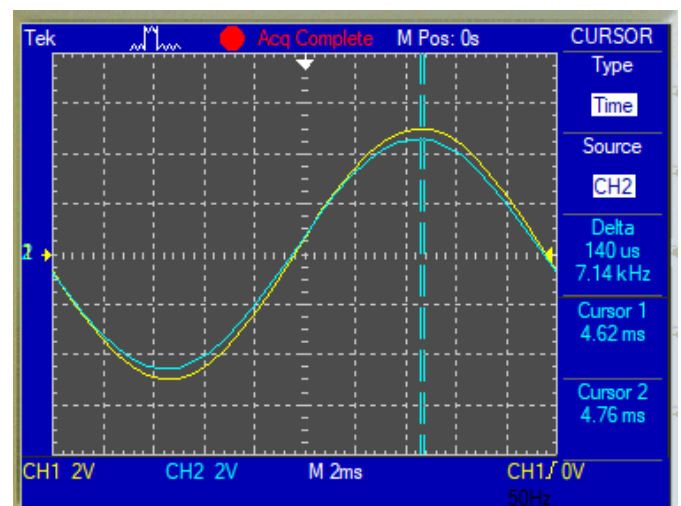
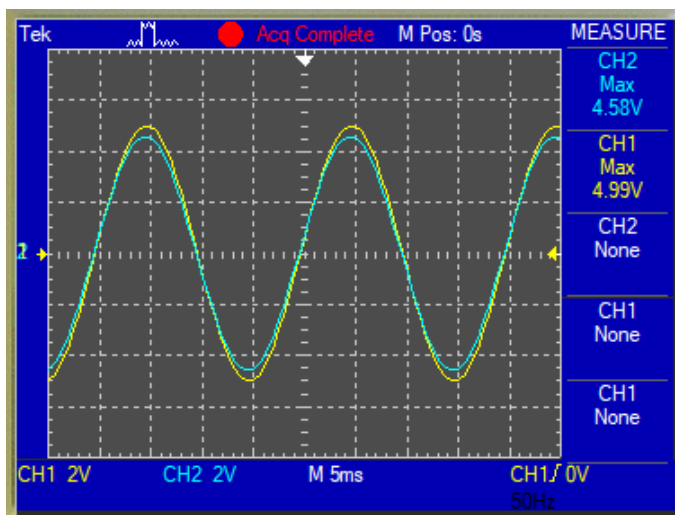
For Inductive load,
 $I_s = \frac{V(R_i)}{R_i}$
 $= \frac{440 \angle -15.84^\circ}{1000}$
 $= 0.44 \angle -15.84^\circ \text{ mA}_{\text{pk}}$
 $\therefore P = \frac{1}{2} \times (I_{\text{pk}})^2 \times R_{\text{load}}$
 $= \frac{1}{2} \times (0.44 \times 10^{-3})^2 \times 10000$
 $= 0.968 \text{ mW}$

Screenshots (Inductive Load):



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Reading 1: Circuit with oscilloscope connection



Problem 2: Capacitive Load in a Series Circuit

Simulate the capacitive load circuit shown in Figure 2. Measure the load voltage, the voltage across R_i , and the required time differences using the oscilloscope. Then complete Table 2 and calculate the load current and power drawn by the load.

Table 2: Capacitive Load

Reading	Parameter	Value
1	V_Load (peak)	4.57 V (peak)
2	Δt for load voltage	80 μ S
3	θ for load voltage	$\theta = (\Delta t \times f \times 360) = (80 \times 10^{-6} \times 50 \times 360) = 1.44^\circ$ (Lagging)
4	V_Ri (peak)	420 mV (peak)
5	Δt for Ri voltage	1.5 mS
6	θ for Ri voltage / current	$\theta = (\Delta t \times f \times 360) = (1.5 \times 10^{-3} \times 50 \times 360) = 27^\circ$ (Leading)
7	$I_s \angle \theta$	$I_s = \frac{V(R_i)}{R_i} = \frac{420 \angle 27^\circ}{1000}$ $= 0.42 \text{ mA}_{pk} \angle 27^\circ$
8	Power drawn by the load, P	$P = \left(\frac{1}{2} I_{pk}^2 \times R_{load}\right) = 0.5 \times (0.42 \times 10^{-3})^2 \times 10^4$ $= 0.882 \text{ mW}$

Calculation (Capacitive Load):

for, capacitive load,

θ for load voltage,

$$\theta = \Delta T \times f \times 360^\circ$$

$$= 80 \times 10^{-6} \times 50 \times 360^\circ$$

$$= 1.44^\circ \text{ (lagging)}$$

θ for Ri voltage / current,

$$\theta = \Delta T \times f \times 360^\circ$$

$$= 1.5 \times 10^{-3} \times 50 \times 360^\circ$$

$$= 27^\circ \text{ (leading)}$$

For capacitive load,

$$I_s = \frac{V(R_i)}{R_i}$$

$$= \frac{420 \angle 27^\circ}{1000}$$

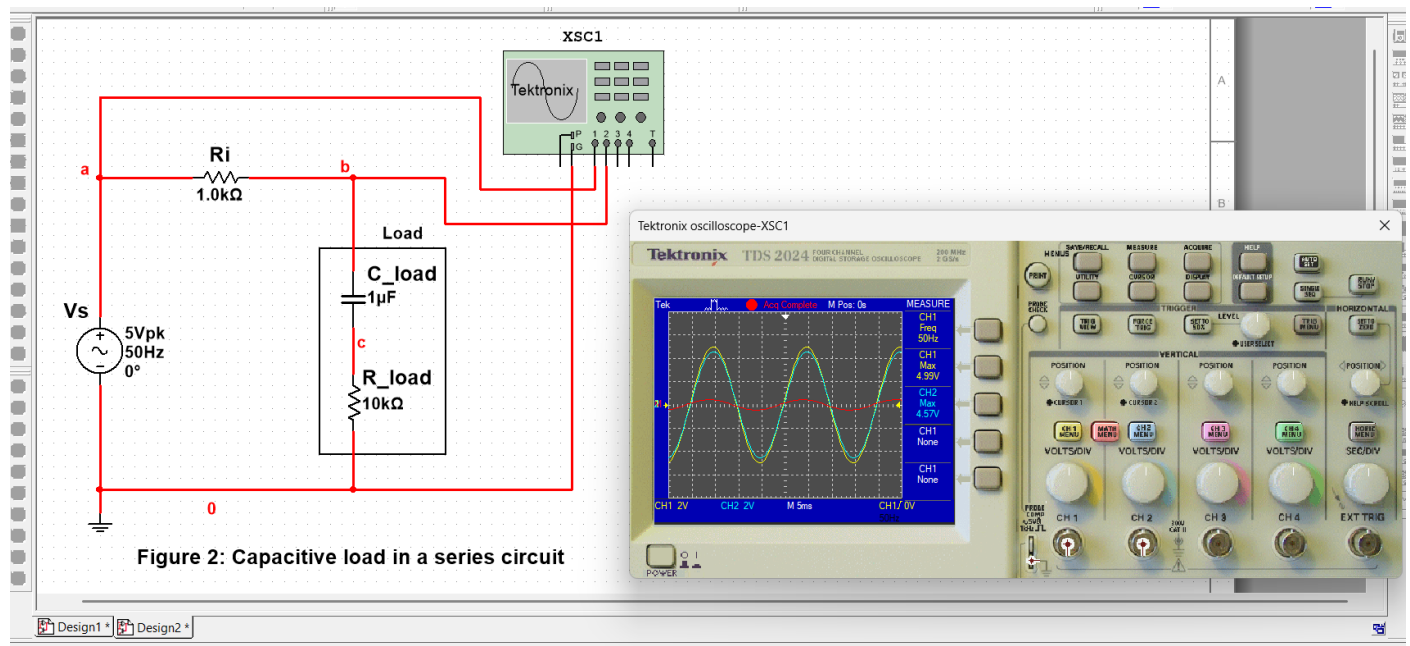
$$= 0.42 \angle 27^\circ \text{ mA}_{pk}$$

$$P = \frac{1}{2} \times (I_{pk})^2 \times R_{load}$$

$$= \frac{1}{2} \times (0.42 \times 10^{-3})^2 \times 10000$$

$$= 0.882 \text{ mW}$$

Screenshots (Capacitive Load):



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Reading 1: Circuit with oscilloscope connection

